Thaddeus Harris | Homework #3 9/14/2024 | Materials for the Built Env.

(3) A 48 inch long aluminum bar (E = 10000 ksi) consists of two, 24 inch long, sections. Section AB is 4.0 inches in diameter, section BC is 2.5 inches in diameter. The bar is loaded in tension by a force, P = 25 kips. Determine total elongation under this load

$$d_{avg} = \frac{4.0in \cdot 24in + 2.5in \cdot 24in}{48in \cdot 2} = \frac{4in + 2.5in}{2in} = 3.25in \quad \sigma = \frac{P}{A} = \frac{25kips}{(3.25in/2)^2 \pi} = 3.014\ kips/in^2 = 3.014\ ksi$$

$$\varepsilon = \frac{\sigma}{E} = \frac{3.014ksi}{10000ksi} = 3.014 \times 10^{-4} = 0.00030$$

(5) Allowable strength = 165 MPa, E = 200 GPa, A = 14.35 cm², l = 4.0 meters

$\sigma_{max} = 165MPa$ $\quad \varepsilon_{max} = \frac{\sigma_{max}}{E}$

$F_{max} = 165MPa \cdot 0.001435m^2 = 0.2368\ MN$ $\quad \varepsilon_{max} = \frac{165MPa}{200GPa} = \frac{165MPa}{200000MPa} = 0.000825$

$U_{max} = \sigma_{max} \cdot \varepsilon_{max} \cdot 0.5$

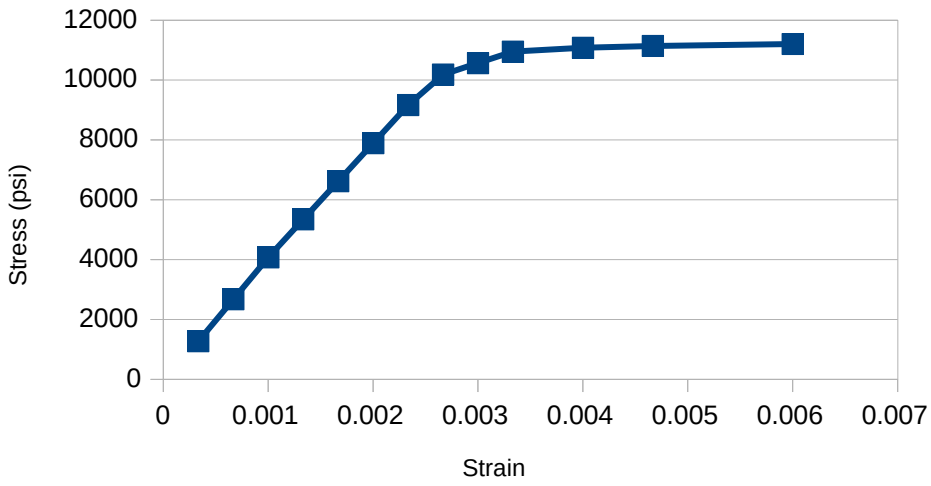
$U_{max} = 165MPa \cdot 0.000825 \cdot 0.5 = 0.0681\ MPa$

$KE$ = Strain Energy = $U_{max} \cdot V$

$KE = 0.1435m^2 \cdot 4.0m \cdot 0.0681MPa$

$KE = 0.0391\ MJ$

$KE = 39100\ J$

| Load (lb) | Elongation (in) | Stress (psi) | Strain | Strain Energy (lb-ft) | Strain Energy Density (psi) |
|---|---|---|---|---|---|
| 0 | 0.000 | 0.00 | 0.00000 | 0.00000 | 0.00000 |
| 40 | 0.001 | 1273.24 | 0.00033 | 0.02000 | 0.21221 |
| 84 | 0.002 | 2673.80 | 0.00067 | 0.08400 | 0.89127 |
| 128 | 0.003 | 4074.37 | 0.00100 | 0.19200 | 2.03718 |
| 168 | 0.004 | 5347.61 | 0.00133 | 0.33600 | 3.56507 |
| 208 | 0.005 | 6620.85 | 0.00167 | 0.52000 | 5.51737 |
| 248 | 0.006 | 7894.09 | 0.00200 | 0.74400 | 7.89409 |
| 288 | 0.007 | 9167.32 | 0.00233 | | |
| 320 | 0.008 | 10185.92 | 0.00267 | | |
| 332 | 0.009 | 10567.89 | 0.00300 | | |
| 344 | 0.010 | 10949.86 | 0.00333 | | |
| 348 | 0.012 | 11077.18 | 0.00400 | | |
| 350 | 0.014 | 11140.85 | 0.00467 | | |
| 352 | 0.018 | 11204.51 | 0.00600 | | |

(c) Stress-Strain Graph